Introduction to statistics

Hypothesis Testing

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Original slides by Joao Lourenço and Rachel Marcone (SIB)

Modified by Marilin Moor (UT)

A possible experiment

1. We have developed a possible new drug for treating the flu. We want to know if Drug A improves recovery compared to a placebo.

2. Biological Null hypothesis: Drug A and placebo have the same effect on recovery.
   Biological Alternative hypothesis: Drug A and placebo have different effects on recovery.

3. How we measure effect? E.g. recovery score (0-100)

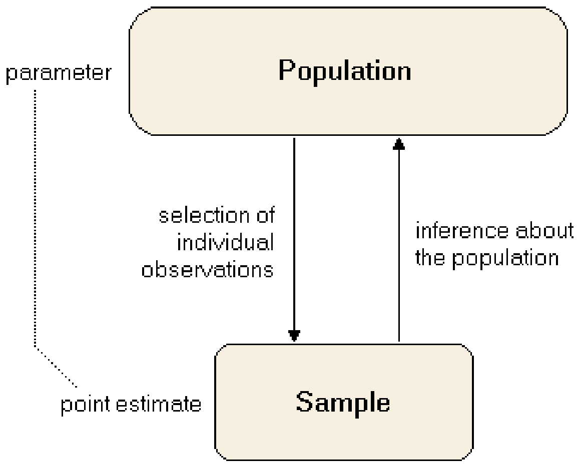
A possible experiment

4. μ_1 = mean recovery score for Drug A and μ_2 = mean recovery score for placebo

Statistical Null hypothesis (H0): $\mu_1 = \mu_2$
Statistical Alternative hypothesis (H1): $\mu_1 \neq \mu_2$

5. Confounding variables? Possibly age of people in the cohort, gender, comorbidity, etc.

*Stats 101 interlude – **population, sample, standard error***



	Sample	Population
Standard deviation	s	σ
Variance	s^2	σ^2
Datapoint	x_i	x_i
Average	$\bar{x}$	μ
Total dataset number	n	N
Standard deviation formula	$s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{\boldsymbol{n-1}}}$	$\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{N}}$
Variance formula	$s^2 = \frac{\sum(x_i - \bar{x})^2}{\boldsymbol{n-1}}$	$\sigma^2 = \frac{\sum(x_i - \mu)^2}{N}$

$$SE = \frac{\sigma}{\sqrt{n}}$$

Standard deviation

Number of samples

A possible experiment (continued)

- We do the experiment, with 800 subjects (number suggested by our power analysis),
  195 were treated with the placebo and 605 with the treatment. For each patient, we record a recovery score after treatment.

- After the study, we obtain Drug A group (n =605), mean recovery score = 78.4, SD = 12.1
- Placebo group (n = 195), mean recovery score = 71.2, SD = 13.0

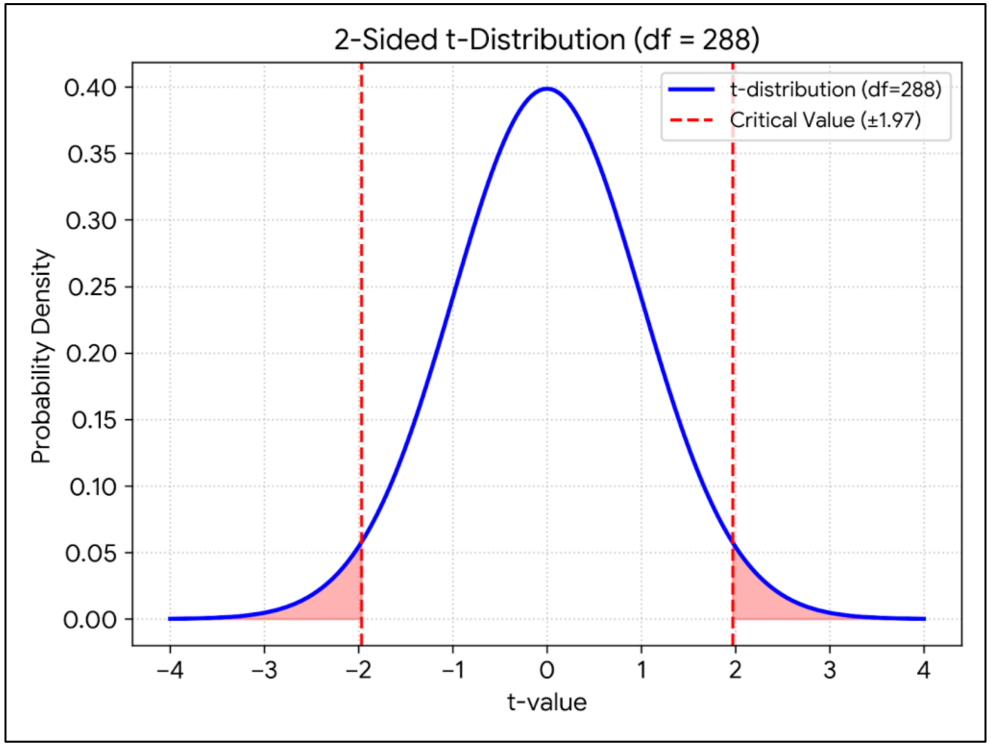
- Is the difference in mean recovery scores statistically significant?

- Comparing means of 2 independent groups → 2-sample t-test

- Choose an alpha/significance/critical level.
  alpha = 0.05

A possible experiment

- Find the critical value equivalent in 2-sample t-test (1.97)
- Found the t-test statistic: 6.84
- Compare the test statistic to the critical value above and decide if you should support or reject the null hypothesis.
- 6.84 > 1.97, so we can reject the null hypothesis.



How to judge whether a difference is significant ?

- The p-value is the probability of getting a result that is as or more extreme than the observed result, assuming that the null hypothesis is true.
- A p-value is **not** the probability that the null hypothesis is correct.
- A p-value is **not** the probability of making an error.

How to judge whether a difference is significant ?

- A predefined significance level (α) is defined, typically 0.05 or 0.01.
- If the observed test statistic is above the threshold, we reject the null hypothesis.
- If "$p<0.05$", we don't know if it is 0.049 (barely significant) or 0.000000001 (extremely significant) - you should report the exact values.
- "$p<0.05$" remains a magical threshold for many scientists.

Difference between two-samples and two-sided tests

- A **two-sample test** is a hypothesis test for answering questions about means for two different populations. Data are collected from two random samples of independent observations.

- A **two-sided test** (or two-tailed test) is a hypothesis test in which the values for rejecting the null hypothesis are in both tails of the probability distribution.

- The choice between a one-sided test and a two-sided test is determined by the purpose of the investigation or prior information.

Two-sided test versus one-sided test

- Two-sided, nondirectional, two-tailed hypothesis tests

  H0: the proportions are the same: A=B

  H1: the proportions are different: A ≠B

  (other verbs like "affect", "change")

- One-sided, directional, upper-tail hypothesis tests

  H0: the proportions are the same: A=B

  H1: A is larger than B: A > B

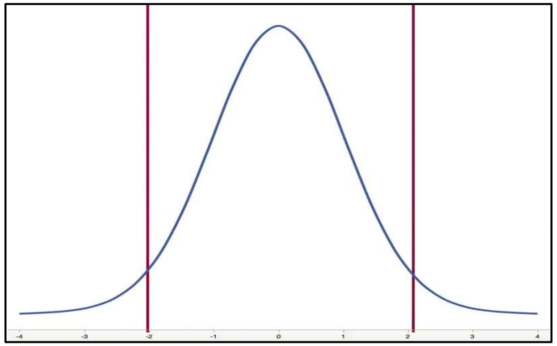
  (other verbs like "increase", "improve")

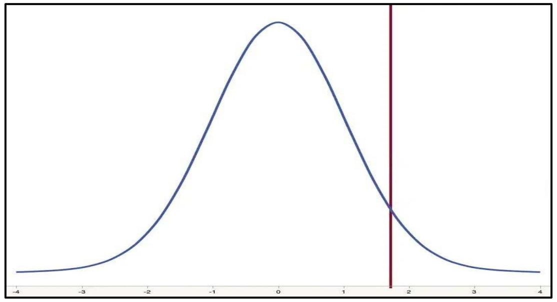
- One-sided, directional, lower-tail hypothesis tests

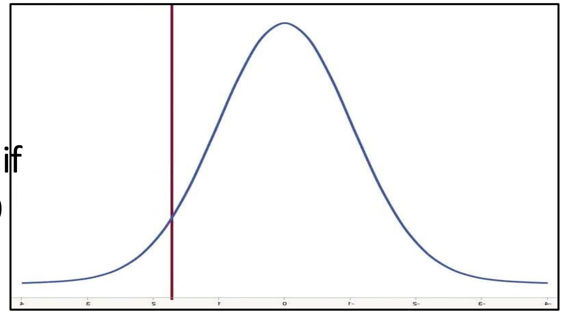
  H0: the proportions are the same: A=B

  H1: A is smaller than B: A < B

  (other verbs like "reduce", "decrease")







Pitfalls in hypothesis testing

- Even if a result is "statistically significant", it can still be due to chance
- Conversely, if a result is not statistically significant, it may be only because you do not have enough data (**lack of power**)
- A test of significance does not say **how important the difference is**, or **what caused it** (Is H0 incorrect? Was an assumption violated Were you unlucky?)
- Using a significance level transforms a complicated, real-world problem, into a simple two-part question

Statistical significance is
not the same
as practical importance.

Published: 14 December 2008

Six new loci associated with body mass index highlight a neuronal influence on body weight regulation

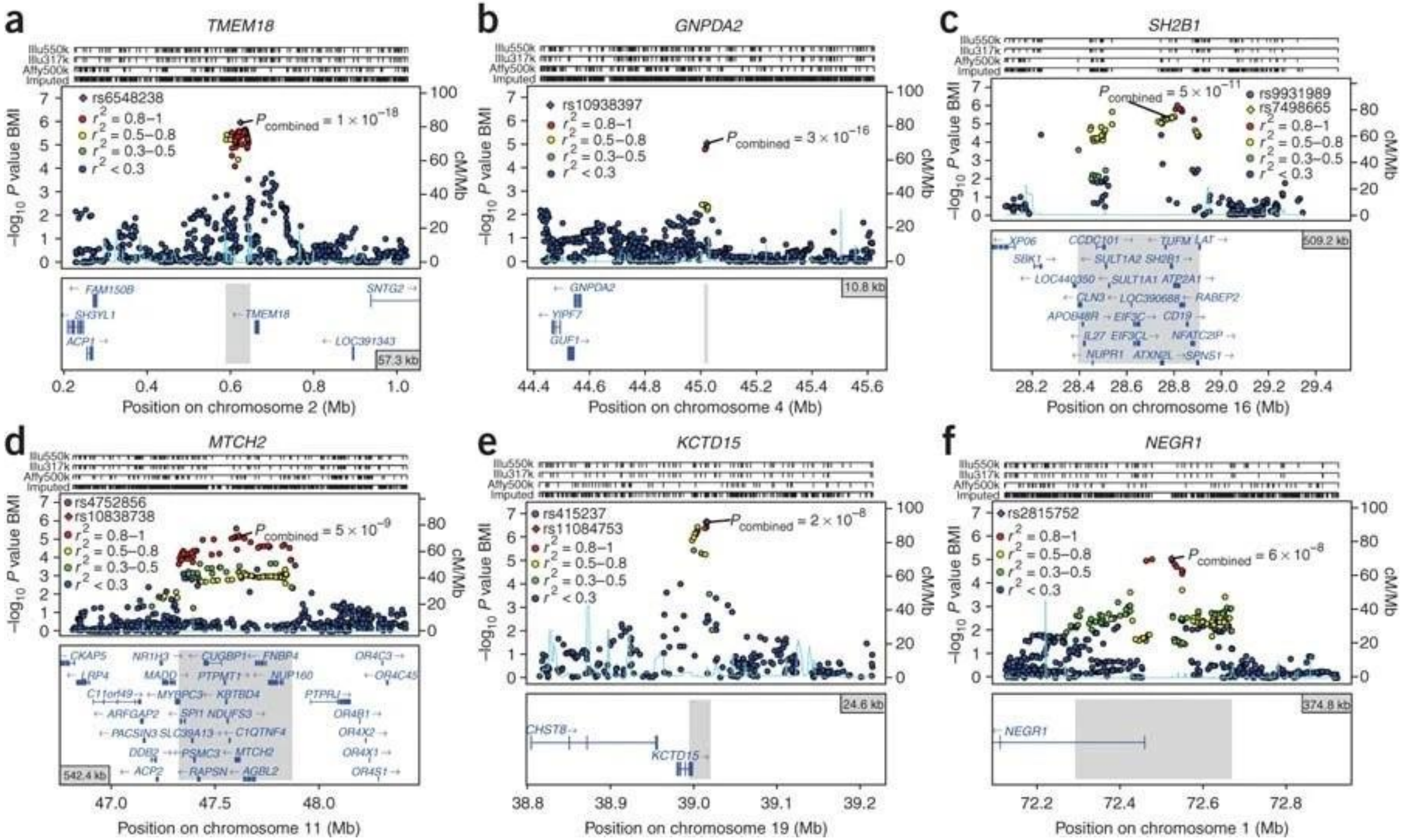
the GIANT Consortium

Nature Genetics **41**, 25–34(2009) | Cite this article

Abstract

Common variants at only two loci, *FTO* and *MC4R*, have been reproducibly associated with body mass index (BMI) in humans. To identify additional loci, we conducted meta-analysis of 15 genome-wide association studies for BMI ($n > 32,000$) and followed up top signals in 14 additional cohorts ($n > 59,000$). We strongly confirm *FTO* and *MC4R* and identify six additional loci ($P < 5 \times 10^{-8}$): *TMEM18, KCTD15, GNPDA2, SH2B1, MTCH2* and *NEGR1* (where a 45-kb deletion polymorphism is a candidate causal variant). Several of the likely causal genes are highly expressed or known to act in the central nervous system (CNS), emphasizing, as in rare monogenic forms of obesity, the role of the CNS in predisposition to obesity.

Statistical significance is not the same as practical importance



8 SNPs (6 discovered in the study) significantly associated with BMI.

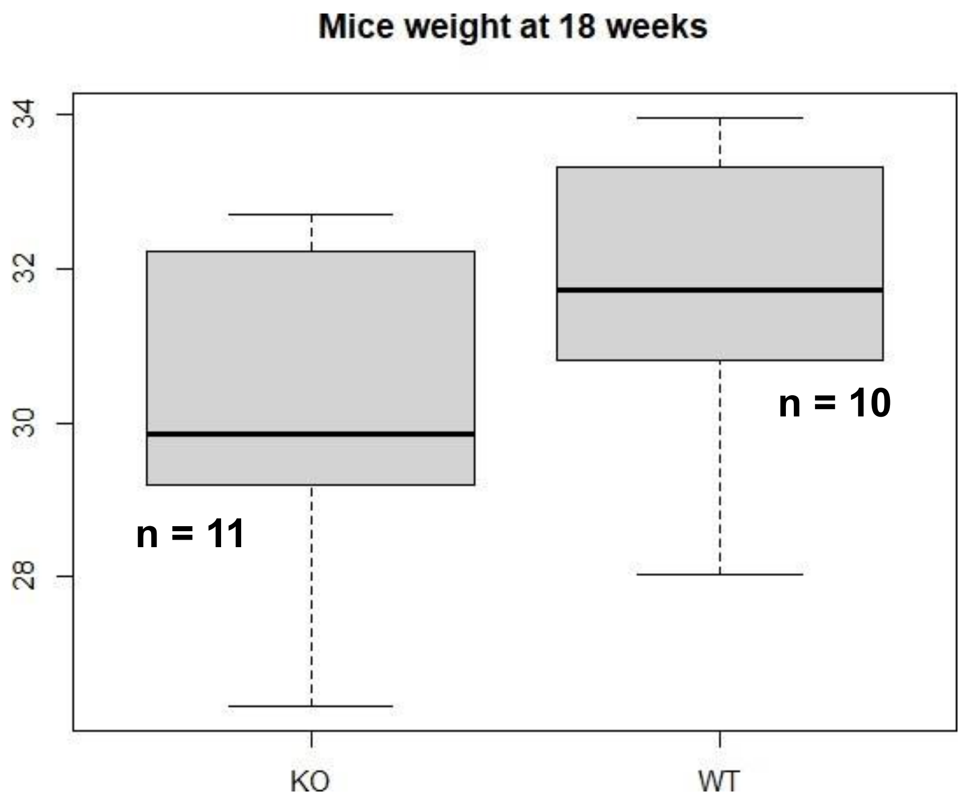
They correspond to a change of **173–954 g** in weight per allele in adults who are 160–180 cm tall

Which test should I use ?

We have data about mice for which a gene was knocked out (KO), as well as control mice (WT)

Question:

Is there a significant difference between the average weight of these two groups ?



Which test should I use ?

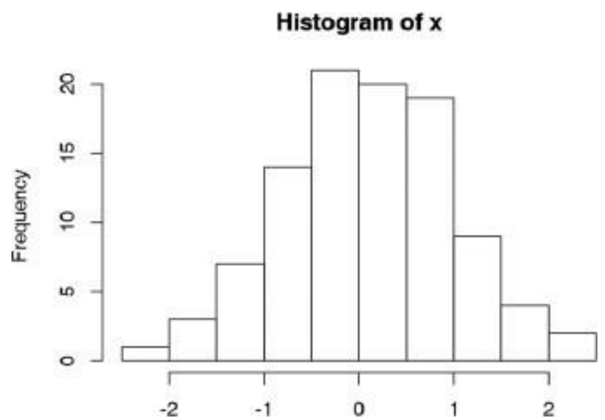
H0: the mean of the two groups is the same
H1: the mean of the two groups is different

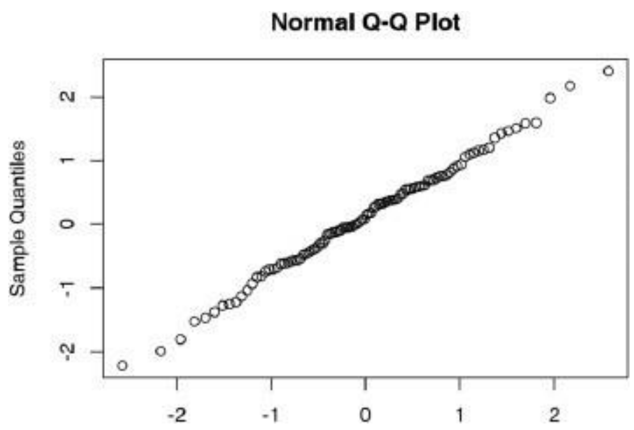
To perform this hypothesis test, we can use a two-sample t-test.

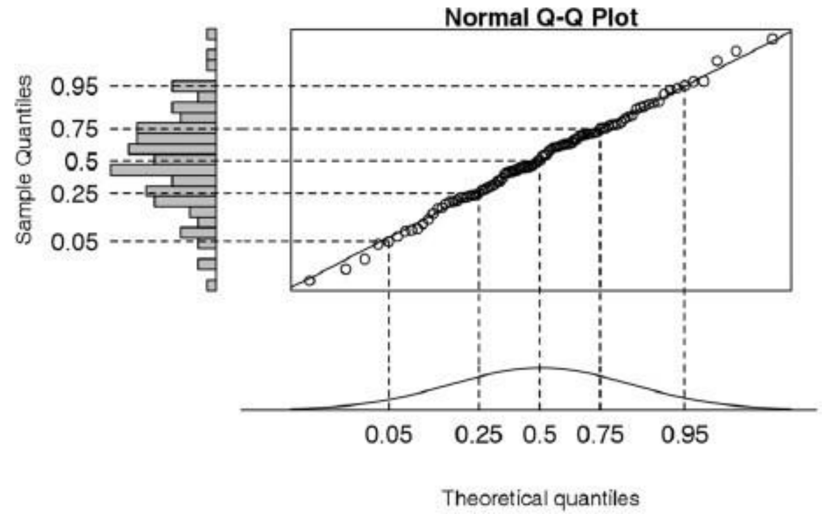
Main assumptions:

- Data values must be **independent**.
- Data in each group must be obtained via a **random sample** from the population.
- Data in each group are **normally** distributed.

How to assess normality?







Example normality tests: Shapiro-Wilk, Kolmogorov-Smirnov:

They test ***against*** normality (and can not show/prove normality)

Normality important to test with sample sizes < 30 (for bigger sizes, the central limit theorem is deemed to take effect)

Which test should I use ?

H0: the mean of the two groups is the same

H1: the mean of the two groups is different

To perform this hypothesis test, we can use a two-sample t-test.

Main assumptions:

- Data values must be **independent**.
- Data in each group must be obtained via a **random sample** from the population.
- Data in each group are **normally** distributed.
- Data values are **continuous**.
- The variances for the two independent groups are **equal (you can use a levene-test, in R : levene_test() to check for this assumption)**.

Two-sample t-test (equal variance)

```
> t.test(KO_WT$weight ~ KO_WT$genotype, var.equal = T)

       Two Sample t-test

data:  KO_WT$weight by KO_WT$genotype
t = -1.4239, df = 19, p-value = 0.1707
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -3.0099018  0.5726089
sample estimates:
mean in group KO mean in group WT 
        30.32366         31.54231 
```

$t = (\bar{x}_1 - \bar{x}_2) / SE$

$\bar{x}_1$ = sample 1 mean (30.32366)

$\bar{x}_2$ =sample 2 mean (31.54231)

SE = standard error of the difference in means (≈ 0.855)

If var.equal = TRUE

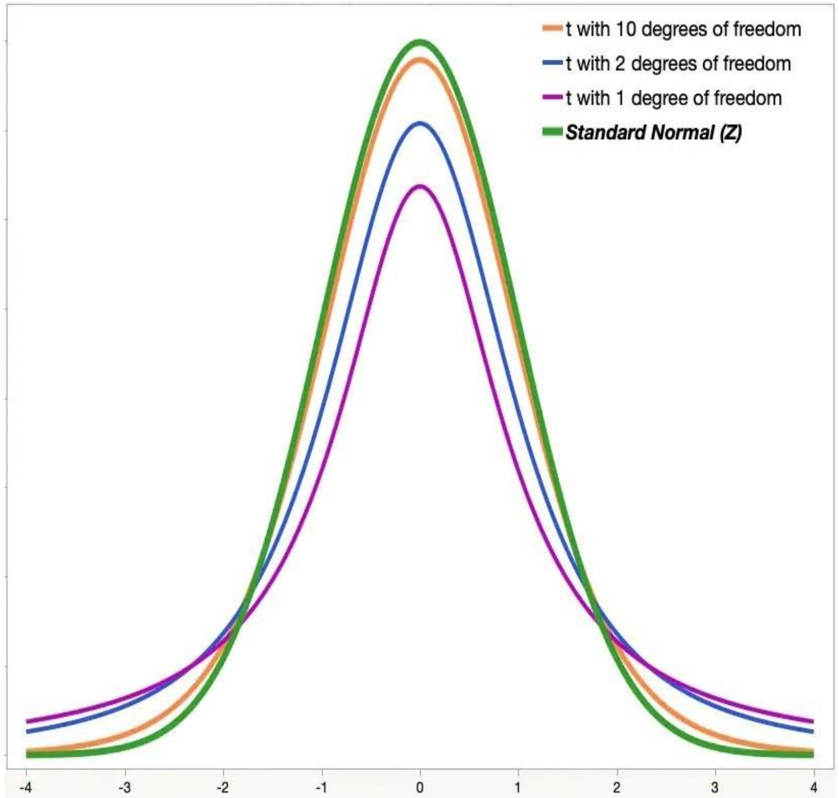
$$df = n_1 + n_2 - 2$$

The t-distribution

The t-distribution describes the standardized distances of sample means to the population mean **when the population standard deviation is not known**, and the observations come from a normally distributed population.

The t-distribution is like a normal distribution:

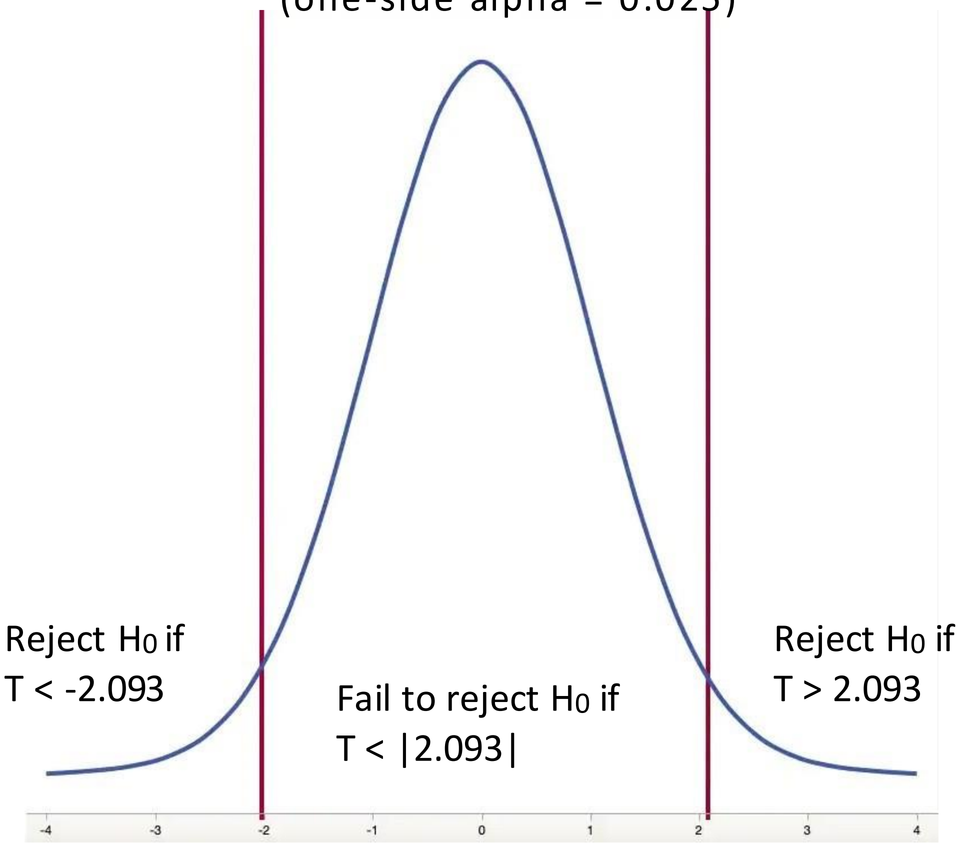
- **Smooth shape.**
- **Is symmetric**.
- **Has a mean of zero**.
- The t-distribution is defined by the **degrees of freedom**
- The t-distribution is most useful for **small sample sizes.**
- **With increasing sample size, converges to the normal distribution**



The t-distribution

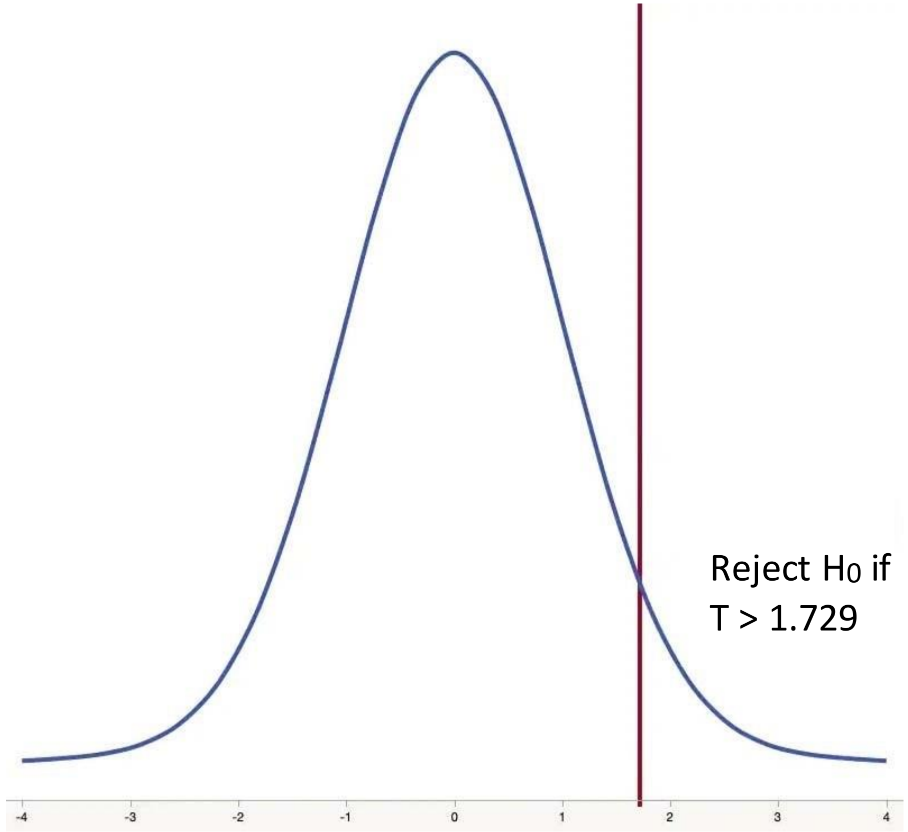
two-tailed test

t-distribution with df = 19, alpha = 0.05
(one-side alpha = 0.025)



one-tailed test

t-distribution with df = 19, alpha = 0.05



Two-sample t-test (equal variance)

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alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -3.0099018  0.5726089
sample estimates:
mean in group KO mean in group WT
        30.32366         31.54231
```

95% CI:
"we are 95% confident the true difference in means (KO-WT) lies between -3.01 and 0.57"

$t = (\bar{x}_1 - \bar{x}_2) / SE$
$\bar{x}_1$ = sample 1 mean (30.32366)
$\bar{x}_2$ = sample 2 mean (31.54231)
SE = standard error of the difference in means (≈ 0.855)
$CI = (\bar{x}_1 - \bar{x}_2) \pm t^* \times SE$
t^* = critical value from t-distribution (2.093 for 2-tailed, 0.95)

Two-sample t-test (equal variance)

```
> t.test(KO_WT$weight ~ KO_WT$genotype, var.equal = T)

        Two Sample t-test

data:  KO_WT$weight by KO_WT$genotype
t = -1.4239, df = 19, p-value = 0.1707
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -3.0099018  0.5726089
sample estimates:
mean in group KO mean in group WT 
        30.32366         31.54231 
```

95% CI:
"we are 95% confident the true difference in means (KO-WT) lies between -3.01 and 0.57"

P-value:
"If the true difference were 0, how often would we see a t-value at least as extreme as -1.4239 just by chance?"

$t = (\bar{x}_1 - \bar{x}_2) / SE$
$\bar{x}_1$ = sample 1 mean (30.32366)
$\bar{x}_2$ = sample 2 mean (31.54231)
SE = standard error of the difference in means (≈ 0.855)
$CI = (\bar{x}_1 - \bar{x}_2) \pm t^* \times SE$
t^* = critical value from t-distribution (2.093 for 2-tailed, 0.95)
p-value = 2 * pt (-abs(t), df = 19))
pt() = cumulative probability, i.e. how much area is to the left of this point t

Two-sample t-test (not equal variance)

```
> t.test(T0_T1$weight_T0,T0_T1$weight_T1)

       Welch Two Sample t-test

data:  T0_T1$weight_T0 and T0_T1$weight_T1
t = -2.3758, df = 23.97, p-value = 0.02585
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -3.8996244 -0.2738217
sample estimates:
mean of x mean of y
 29.89671  31.98343
```

Which test should I use ?

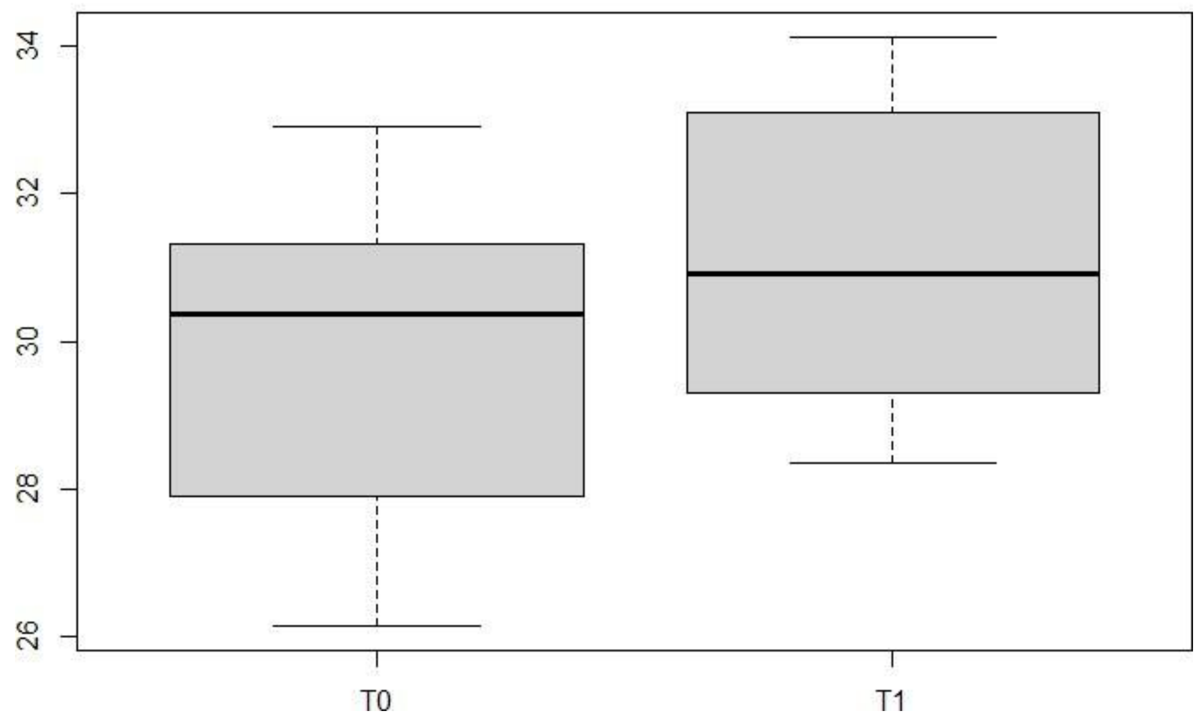
We have data about mice at two different time points (T_0 and T_1)

Question:

Is there a significant difference between the average weight of these mice at these two time points ?

Mice weight at 18 weeks (T0) and at 19 weeks (T1)

n = 13



Which test should I use ?

H0: the mean of the differences is zero
H1: the mean of the differences is not zero

To perform this hypothesis test, we can use a paired t-test.

Main assumptions:

- Subjects must be **independent**. Measurements for one subject do not affect measurements for any other subject.
- Each of the **paired measurements** must be obtained from the **same** subject.
- The measured differences are **normally** distributed.

2-sample t-test vs paired t-test

- In the two-sample t-test, we compared two samples of unrelated data points
- If the data between the two samples is paired, that is, each point x_i in the first sample correspond to a point y_i in the second sample, we can do a paired t-test

Paired t-test

```
> t.test(T0_T1$weight_T0,T0_T1$weight_T1, paired = T)

        Paired t-test

data:  T0_T1$weight_T0 and T0_T1$weight_T1
t = -11.537, df = 12, p-value = 7.491e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -2.480824 -1.692622
sample estimates:
mean of the differences
              -2.086723
```

$$T = \frac{\bar{X} - \bar{Y}}{S \;/\sqrt{n}} \qquad S = \sqrt{\frac{\sum(x_i - y_i)^2 - \frac{(\sum(x_i - y_i))^2}{n}}{n-1}} \qquad df = n - 1$$